

Thermal Extremes and first hospitalisation after coronary heart disease or heart failure diagnosis in Hong Kong, 2013–2023

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Abstract

Background. Hong Kong records more hot nights than a decade ago, while cold days persist. Whether the cold-dominant pattern of earlier daily cardiac-admission studies extends to monthly first-hospitalisation counts is untested.

Methods. We analysed 132 territory-months (2013–2023) of monthly counts of first hospitalisation after a first diagnosis of coronary heart disease (CHD; 156,156 events) or heart failure (HF; 29,681 events) among people with type 2 diabetes and/or hypertension. Admission cause was not recorded. Three continuous temperature measures and official counts of hot nights, very hot days, and cold days each entered a separate negative-binomial model with calendar-month indicators, a 4-df time spline, and a days-in-month offset. Intervals used model-based, HC1, and Newey–West lag-3 and lag-6 constructions. Benjamini–Hochberg q-values covered the twelve exploratory contrasts. A method for recovering daily exposure coefficients from monthly sums failed simulation calibration and was not applied to the health series.

Results. Under Newey–West lag-6 reporting, the count ratio was 1.022 (1.002–1.042) per five hot nights for CHD and 1.073 (1.006–1.144) per five cold days for HF, both with $q = 0.192$; all twelve q-values exceeded 0.19. For CHD hot nights, model-based and HC1 intervals included 1, as did the pre-2020 estimate (1.011, 0.991–1.032). For HF cold days, all four constructions excluded 1, and the pre-2020 estimate was stronger (1.113, 1.053–1.176). Official cold days fell almost entirely in December–February (141 of 145), identifying that contrast between winters.

Conclusions. The data do not support a multiplicity-protected differential thermal claim. The HF cold-day association is concordant across standard-error methods; the CHD hot-night association depends on the construction and on including 2020–2023. Estimates are ecological monthly count ratios for a first-event outcome without recorded admission cause.

Keywords: hot nights; cold days; coronary heart disease; heart failure; Hong Kong; monthly time series

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39 Introduction

40 Temperature-related cardiovascular morbidity is an important concern in subtropical cities,
41 where populations may experience both intense humid heat and episodic cold despite generally
42 warm climates. Acute myocardial infarction and stroke may respond differently to thermal exposure,
43 and the observed burden can vary according to age, housing, behavioral adaptation, air pollution,
44 and access to temperature-controlled environments. These questions are especially relevant in Hong
45 Kong, a densely populated and rapidly aging city with substantial seasonal variation and a growing
46 frequency of hot nights and very hot days.

47 Earlier Hong Kong evidence emphasized cold-related cardiovascular risk. Using daily data
48 from 2000–2009, Goggins et al. reported that decreases in temperature below an estimated 24°C
49 threshold were associated with increased AMI admissions, while no statistically significant heat
50 effect was observed [1]. The same study identified nitrogen dioxide as an important predictor of AMI
51 admissions.

52 Hong Kong's thermal environment has changed since these historical study periods. At the
53 Hong Kong Observatory (HKO), the number of hot nights increased from 10 in 2013 to 61 in 2021,
54 while very hot days increased from 17 to 54 in the same selected-year comparison [2, 4]. Cold days
55 nevertheless continued to occur, indicating that contemporary preparedness must consider
56 potentially overlapping heat and cold hazards.

57 Demographic change may further alter the distribution of cardiovascular burden. Age-sex-
58 specific population denominators and appropriate person-time offsets are necessary to distinguish
59 changing population structure from changes in admission rates. The environmental context has also
60 evolved: temperature and pollutants should not be treated as statistically interchangeable
61 covariates, and ozone in particular may act as a confounder, effect modifier, or pathway variable.

62 This study will examine monthly AMI, ischemic stroke, and hemorrhagic stroke admissions
63 among Hong Kong residents aged 35 years or older from 2013 to 2023. It will assess associations
64 with HKO-defined thermal extremes, emphasizing hot nights and cold days, while accounting for
65 population denominators, seasonality, long-term trends, and COVID-period disruption. The objective
66 is not to assume that heat has replaced cold, but to determine whether the historically cold-
67 dominant pattern remains sufficient to describe contemporary cardiovascular hospital burden in a
68 warming and aging Hong Kong.

70 Methods

71 Study design

Commented [H3]: Remove now since the scope of the study has changed?

Commented [H4]: You need to cite the papers for these factors. In the context of this study, if we don't have the housing and behavioural adaptation data, I would suggest to not mention it and instead replace it with medication instead. You don't want to encourage people to think about a possible factor that we did not control for from the get-go.

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72 The design is an ecological territory-month time series covering January 2013 to December
 73 2023. For each single-exposure model, the **target quantity** is a count ratio associated with a 1°C or
 74 five-day exposure contrast, conditional on calendar month and a smooth function of time. Individual
 75 causal effects, principal-diagnosis CHD or HF, acute myocardial infarction, and daily triggering are not
 76 identified [14]. Monthly counts of cohort members still at risk of a first event were unavailable, so the
 77 offset is **days** in month and the estimates are monthly count ratios rather than cohort incidence-rate
 78 ratios [14]. The twelve-contrast model set was specified after the outcome series were available and
 79 is exploratory. We refer to the twelve separate single-exposure models as the core panel. In line with
 80 ethical requirements, the study was approved by the Institutional Review Board (IRB) of the
 81 HKU/Hospital Authority HK West Cluster (IRB reference number: UW XX-XXX).

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Commented [KW8]: Do you mean "the number of days in a month"? This would make it more clear.

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Commented [KW10]: @Bob: You only need one sentence here to indicate that this study has ethics approval. All the other stuff is not required (I deleted it already).

Commented [KW11]: @Roro: Can you input the IRB reference number here for this study?

82 83 *Health data*

84 Governed monthly aggregates were constructed and released by the outcome co-
 85 investigator. The cohort comprised people diagnosed with type 2 diabetes and/or hypertension
 86 during 2013–2023. For each outcome, the event was the first recorded hospitalisation after the
 87 patient's first diagnosis record for CHD or HF. Diagnosis-record construction was assisted by a clinical
 88 collaborator. Admission cause was not recorded. Delivered columns were labelled **chd_inpatient** and
 89 **hf_inpatient** at territory × calendar month grain. Inpatient-only semantics are inferred from those
 90 labels and remain subject to confirmation. Age–sex strata were not included. A stroke series was
 91 named in correspondence but was not attached; no stroke analysis is reported. Field-level ICD lists
 92 and the executable timing rule remain with the outcome co-investigator and should replace this
 93 paragraph where they differ from it.

94 Monthly influenza activity between November 2013 and December 2023 was obtained from
 95 Centre for Health Protection Flu Express summaries. For monthly influenza activity data that was
 96 unavailable between January 2013 and October 2013, data points from these months were considered
 97 to be missing values and were not coded as zero. Influenza did not enter the core panel.

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Commented [KW13]: Brief justification needed.

98 99 *Weather and pollutants data*

100 Meteorological data was obtained from the HKO. The monthly variables included:
 101 temperature (mean, minimum, maximum), **relative humidity**, **total rainfall**, the number of hot nights
 102 ($T_{\min} \geq 28^{\circ}\text{C}$), the number of very hot days ($T_{\max} \geq 33^{\circ}\text{C}$), the number of extremely hot days ($T_{\max} \geq$
 103 35°C), and the number of cold days ($T_{\min} \leq 12^{\circ}\text{C}$). Monthly mean pollutant levels for nitrogen dioxide
 104 (NO_2), sulfur dioxide (SO_2), and ozone (O_3) were obtained from the Hong Kong Environmental

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105 Protection Department (EPD). All monthly data were derived by taking the average of daily data in
106 each calendar month.

107

108 Core exposures were monthly mean temperature, the monthly means of daily maximum
109 and minimum temperature, and official counts of hot nights, very hot days, and cold days. The
110 three counts were scaled per five days, so that a coefficient corresponds to a five-day contrast
111 rather than to a single extreme day.

112 Official cold days were concentrated in December–February (Results, Figure 2), so
113 with calendar-month indicators in the model the remaining cold-day variation is a difference
114 between years within winter months. A winter-only restriction is a labelled sensitivity, and not
115 a remedy for that identification. Extremely hot days, relative humidity, rainfall, and sulfur
116 dioxide were assembled with the weather series but did not enter the core panel.

117 Fine particulate matter (PM2.5) from general EPD stations was assembled for staged
118 pollution sensitivities. A station-month contributed to a pollutant mean only when at least 75%
119 of expected observations were available, and missing measurements were not coded as zero.
120 Territory-wide monthly concentrations were unweighted means across eligible general
121 stations. Roadside stations were reserved. Nitrogen dioxide and particulate matter were
122 specified to enter before ozone, because ozone is coupled with hot, sunny conditions. Staged
123 pollution models exist in the archive; they are not adjusted versions of the core panel, and they
124 were not used to claim that confounding is resolved. Operational heatwave-spell and hot-month
125 rules beyond the official daily thresholds remain with the weather co-investigator, and
126 provisional monthly-tail indicators were not used as confirmatory exposures.

127 Population denominators and offsets

128 Age- and sex-specific mid-year population estimates from the Census and Statistics
129 Department (Table 110-01001) were linearly interpolated to calendar-month midpoints [8].
130 Those denominators describe the general population, and not the diabetes or hypertension
131 cohort still at risk of a first CHD or HF hospitalisation. Core models therefore used a days-in-
132 month offset, $\log(dt)$, and are interpreted as monthly count ratios [14]. A general-population $\times$
133 days offset was retained only as a sensitivity. It does not convert the estimand into cohort
134 incidence.

135

136 *Statistical analysis*

Commented [KW15]: Why five days and not three days?
How do the values differ if you used a single extreme day?
Maybe you can do a sensitivity analysis to check?

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meaning it goes in the Results.

Commented [KW17]: What does this mean? Reserved
for what?

Let Y_t be the monthly count in month t . Core models used a negative-binomial likelihood [10]: $\log E(Y_t) = \log(dt) + \alpha + \beta X_t + \text{month indicators} + s(t; 4 \text{ df})$, where dt is days in month, X_t is a single exposure, and $s(t; 4 \text{ df})$ is a natural cubic spline of month index. Each core model entered one exposure at a time: mean temperature, mean maximum temperature, mean minimum temperature, hot nights, cold days, or very hot days, for CHD and for HF. That set defines the twelve core contrasts. Quasi-Poisson models were a family sensitivity. Negative binomial was preferred because monthly counts were overdispersed relative to Poisson.

Core intervals used Newey–West standard errors with lag 6 [19], reported together with model-based, HC1, and Newey–West lag-3 intervals, treating the fitted negative-binomial score equations as estimating equations. Lag 6 was the core reporting choice because residual serial dependence is expected in monthly counts and is material for CHD. Robust intervals are not automatically wider than model-based intervals, and the four-construction ladder is reported for that reason. No interval was selected because it excluded 1.

Joint models that entered maximum and minimum temperature together, or the three official extreme-day counts together, were retained as collinearity diagnostics. Variance inflation factors were computed after residualising on calendar month and the trend spline.

Robustness analyses comprised alternative trend smooths (3, 6, and 8 df), year fixed effects, exclusion of the first 12 or 24 months, a pre-2020 window (January 2013–December 2019; 84 months), COVID-phase adjustment, lag-0, lag-1, and lag-2 month exposures, and exclusion of the month with largest Cook’s distance. COVID phases were pre-COVID (through January 2020), early COVID (February 2020–December 2021), fifth wave (January–April 2022), late 2022 (May–December 2022), and post-reopening (from January 2023). Pearson residual autocorrelation and Ljung–Box tests at lags 6 and 12 were inspected. Benjamini–Hochberg q -values were computed across the twelve core contrasts [20].

Archive models that entered pollution, humidity, or influenza are not adjusted versions of the core panel. They were not used to claim that confounding is resolved. **Constrained daily-exposure recovery**

A constrained method for recovering daily exposure coefficients from monthly sums was evaluated in simulation against Hong Kong weather, following the Basagaña–Ballester aggregation framework [15,16]. Implementation was from the published estimating equation. Public code without a repository licence was not copied. Synthetic daily outcomes were matched to the disclosure-minimised seasonality, scale, overdispersion, and residual

dependence of the monthly series. Admission of any real daily coefficient required all frozen calibration criteria to pass, including Type I error, coverage, relative bias, and sign recovery. A 500-replicate calibration against Hong Kong weather failed those requirements. A subsequent worst-cell re-summary of the same fits did not reverse that decision. No daily health coefficient was produced. This is a project-specific calibration result, and it does not show that the Basagaña–Ballester method fails in general. A full daily distributed-lag non-linear model is not identified from 132 monthly sums.

Software

Analyses were conducted in R 4.3.3 (2024-02-29). Negative-binomial models used MASS::glm.nb (MASS 7.3-60.0.1). Newey–West and HC1 standard errors used sandwich 3.1.3. Natural cubic splines used splines::ns.

Commented [KW18]: Good.

Results

Outcome series

CHD contributed 156,156 first recorded hospitalisations over 132 months, a mean of 1,183.0 per month. HF contributed 29,681, a mean of 224.9 per month (Table 1). Annual totals fell by about half from 2013 to 2023 (CHD 23,830 to 12,323; HF 4,336 to 2,296), with a further dip in 2020 (CHD 10,237; HF 1,964). Over the same years, the interpolated Census and Statistics Department population aged 35 years or older rose by 17%. Figure 1 indexes both series to 2013. The decline in first-event counts is compatible with depletion of the at-risk set in a 2013–2023 diagnosis window, together with pandemic changes in care-seeking. It is not a measure of falling incidence in the territory, and it does not quantify the share attributable to depletion without monthly still-at-risk person-time. Mean monthly HF counts were highest in January (287) and lowest in September (196). CHD showed a milder winter elevation (January 1,386; September 1,097). After calendar-month indicators, the thermal estimates ask whether a given January or July differs from a typical January or July, and not whether winter counts exceed summer counts.

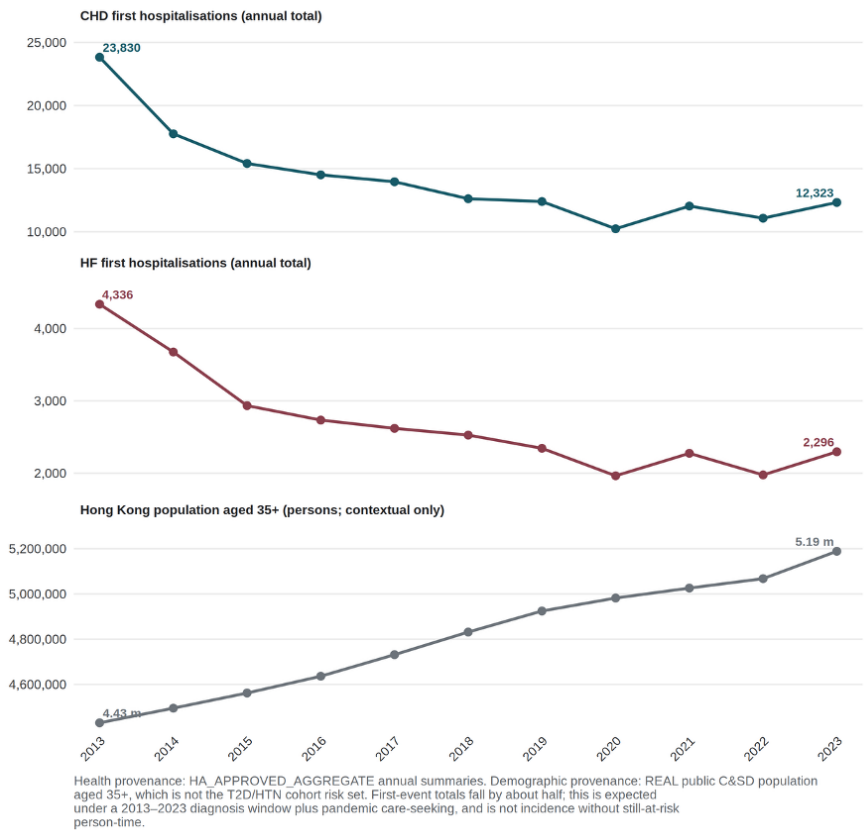
Table 1. Outcome summary.

Outcome	Period	Months	Total events	Mean per month	Event construction
Coronary heart disease	2013–2023	132	156,156	1,183.0	First recorded hospitalisation

					after first CHD diagnosis
Heart failure	2013–2023	132	29,681	224.9	First recorded hospitalisation after first HF diagnosis

First-hospitalisation totals decline while the broad population rises

2013–2023 change: CHD -48%, HF -47%, Hong Kong population aged 35+ +17%



197

198 Exposure context

199 At HKO Headquarters, hot nights rose from 10 in 2013 to 61 in 2021 and 56 in 2023
200 [6]. The 2013 count was about seven days below the 1981–2010 normal [2]. The 2021 count
201 was the highest annual number of hot nights in the Observatory record at that time [4]. Very
202 hot days rose from 17 to 54 over the same 2013–2021 comparison, and were 54 in 2023 [6].
203 Cold days were 14 in 2013 and 14 in 2023, with 1 in 2019 and 13 in 2021 [6]. These are
204 exposure counts, and not health effects. Figure 2 shows official cold days by year and month.

Of 145 cold days in 2013–2023, 141 fell in December–February (December 40, January 54, February 47), and 4 fell in March. Twenty-nine of 132 months carried at least one official cold day. The year 2019 contributed a single cold day. After calendar-month indicators, the remaining cold-day variation is between-year winter variation, and not a summer-versus-winter contrast.

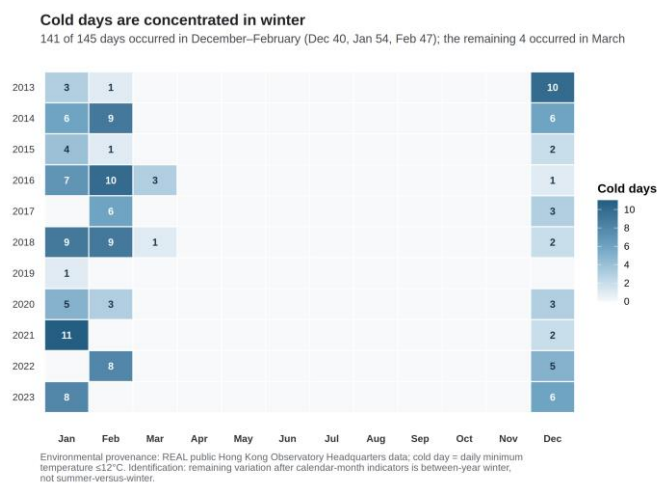


Figure 2. Official cold days by year and month, Hong Kong Observatory Headquarters, 2013–2023. Source: Hong Kong Observatory official daily cold-day flags ($T_{\min} \leq 12^{\circ}\text{C}$), aggregated to calendar months.

Core panel

Table 2 reports the twelve separate-exposure count ratios under the days-in-month offset and Newey–West lag-6 intervals.

Table 2. Separate negative-binomial models with days-in-month offset and Newey–West lag-6 intervals.

Outcome	Exposure contrast	Count ratio (95% CI)	p	BH q
CHD	Mean temperature / 1°C	0.993 (0.976–1.010)	0.424	0.726
CHD	Mean maximum temperature / 1°C	0.993 (0.979–1.007)	0.321	0.642
CHD	Mean minimum temperature / 1°C	0.994 (0.977–1.012)	0.509	0.764

CHD	Hot nights / 5 days	1.022 (1.002–1.042)	0.032	0.192
CHD	Cold days / 5 days	0.995 (0.949–1.043)	0.823	0.900
CHD	Very hot days / 5 days	0.999 (0.974–1.025)	0.962	0.962
HF	Mean temperature / 1 °C	0.974 (0.947–1.002)	0.069	0.207
HF	Mean maximum temperature / 1 °C	0.981 (0.958–1.005)	0.113	0.272
HF	Mean minimum temperature / 1 °C	0.973 (0.947–1.000)	0.050	0.202
HF	Hot nights / 5 days	1.003 (0.976–1.031)	0.825	0.900
HF	Cold days / 5 days	1.073 (1.006–1.144)	0.031	0.192
HF	Very hot days / 5 days	0.995 (0.963–1.028)	0.764	0.900

All twelve q-values exceeded 0.19, with a minimum of 0.192. No contrast meets a multiplicity-protected confirmatory threshold in this exploratory panel. The HF mean-minimum-temperature interval narrowly included 1, with an unrounded upper bound of 1.00005.

For CHD, the continuous monthly temperature estimates were near null, between 0.993 and 0.994, whereas the official hot-night count estimate was larger under Newey–West lag-6 reporting. These encodings describe different monthly thermal questions. The contrast does not imply that a hot-night effect was missed by the means, and it is not a test of hourly nighttime excess heat [17]. For HF, the continuous temperature estimates were weakly inverse, and the cold-day count estimate was positive.

Uncertainty ladder

Table 3. Uncertainty ladder for the two leading exploratory contrasts.

Outcome and exposure	Model	HC1	NW3	NW6
CHD hot nights / 5 days	1.022 (0.995–1.049)	1.022 (0.997–1.047)	1.022 (1.0003–1.0439)	1.022 (1.002–1.042)
HF cold days / 5 days	1.073 (1.023–1.125)	1.073 (1.011–1.138)	1.073 (1.007–1.143)	1.073 (1.006–1.144)

For CHD hot nights, the model-based and HC1 intervals included 1, while both Newey–West intervals excluded 1 on the unrounded scale (lag 3: 1.000253 to 1.043860). The ladder

narrowed from the model-based interval to the Newey–West intervals for that contrast. Robust intervals are not automatically wider, and the direction of that change is itself informative. For HF cold days, all four constructions excluded 1, and q remained 0.192. Concordance across standard-error methods does not create multiplicity protection.

Joint models and residual diagnostics

After month and trend residualisation, maximum and minimum temperature retained a variance inflation factor of 4.66 when entered jointly. Hot nights and very hot days retained variance inflation factors near 1.96. In the joint extreme-day model, the CHD hot-night count ratio was 1.045 (1.015–1.075), compared with 1.022 in the separate model. The HF cold-day estimate was the same in the joint and separate models, at 1.073. Joint coefficients are diagnostics, and not preferred estimates.

Pearson residual autocorrelation at lag 1 was 0.508 in the CHD hot-night model and 0.146 in the HF cold-day model (Supplementary Figure S1). Ljung–Box tests at lag 6 gave $p < 10^{-7}$ for every CHD core model and $p > 0.3$ for every HF core model. Serial dependence is part of the inferential problem for CHD, which is why the four-construction ladder is shown rather than a single interval.

Sensitivity analyses

Offset choice changed the core count ratios only trivially. Across trend, window, and COVID-phase specifications, the CHD hot-night ratio ranged from 1.011 to 1.025, and the HF cold-day ratio from 1.043 to 1.113 (Figure 3). The HF cold-day association was strongest before 2020 (1.113, 1.053–1.176). The CHD hot-night association was weaker and compatible with 1 in the pre-2020 window (1.011, 0.991–1.032) and under COVID-phase adjustment (1.013, 0.994–1.033). The decline in first events and the rise in hot nights are both strong trends across this window, and the same 4-df spline absorbs both. Lag-1 month models attenuated the CHD hot-night association toward 1 (1.010, 0.979–1.042). The HF cold-day association remained elevated at lag 1 (1.073, 1.014–1.135) and was weaker at lag 2 (1.053, 0.985–1.127). Lag-one month is a labelled sensitivity, and it is not a daily lag curve. The months with largest Cook’s distance were February 2020 for CHD and February 2022 for HF cold days. Excluding the most influential month left both exploratory directions unchanged (CHD hot nights 1.021, 1.001–1.040; HF cold days 1.088, 1.034–1.144).

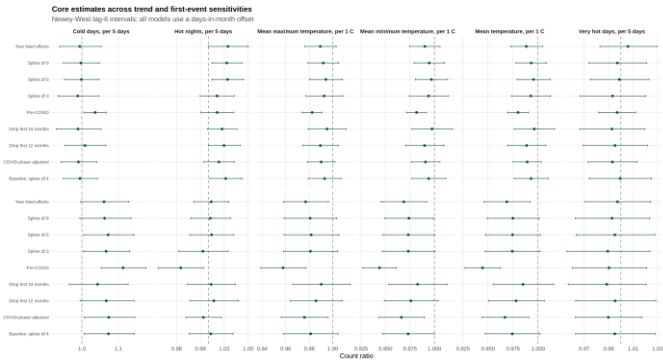


Figure 3. Core count ratios across trend, window, and COVID-phase specifications (Newey–West lag-6 intervals). All twelve core contrasts are shown across nine specifications. The pre-2020 CHD hot-night interval includes 1, while the pre-2020 HF cold-day interval strengthens. Source: sensitivity estimates computed from the governed aggregate series (trend and depletion sensitivity table).

The simulated daily-recovery method failed its worst-cell criteria: null Type I error ranged from 0.048 to 0.150, minimum coverage was 0.840, maximum non-null relative bias was 32.8, and the maximum moderate false-sign rate was 0.808. No real daily coefficient is reported. That refusal is specific to this monthly series and this implementation.

Discussion

These data do not support a multiplicity-protected differential thermal claim for CHD relative to HF. Under Newey–West lag-6 reporting for the core panel, CHD first-hospitalisation counts were more closely associated with official hot-night burden than with mean temperature or cold days. HF counts were more closely associated with cold-day burden than with hot nights. Both patterns sit inside a twelve-contrast family in which every q-value exceeds 0.19. The complete panel is therefore reported, and no contrast is promoted to a primary result.

The HF cold-day association is the more coherent of the two residual signals. It is concordant across all four standard-error constructions. It survives exclusion of the most influential pandemic month. It is not produced by entering correlated heat metrics jointly. It is strongest in the pre-2020 window, before pandemic disruption of care-seeking. It nevertheless remains unprotected by its q-value. Because official cold days fall almost entirely in December–February, and because only 29 of 132 months carry any official cold day, the

290 association is identified from differences between winters rather than from a summer-versus-
291 winter contrast.

292 The CHD hot-night association is smaller, depends on the uncertainty method, and was
293 not evident in the pre-2020 window. Model-based and HC1 intervals include 1, whereas the
294 Newey–West intervals exclude 1 on the unrounded scale (lag 3: 1.000253 to 1.043860).
295 Pearson residual autocorrelation at lag 1 is 0.508 in that model, so inference for the CHD series
296 depends on how serial dependence is handled. For this contrast the Newey–West intervals were
297 narrower than the model-based interval, which is atypical under positive residual
298 autocorrelation; the exclusion of 1 therefore rests on the smaller robust variance estimate, and
299 is a reason for caution rather than confirmation. No interval construction was chosen because
300 it excluded 1. The estimate is also confined to specifications that include 2020–2023, which
301 are the years in which hot nights peaked and in which care-seeking changed. Hot-night burden
302 and the decline in first events both trend strongly across the window, and both are controlled
303 by the same trend spline, so the hot-night contrast carries the harder identification problem of
304 the two.

305 The direction of the HF cold-day residual is consistent with earlier daily evidence that
306 lower temperature was associated with higher heart-failure admissions in Hong Kong [21]. The
307 comparison is between questions, and not between magnitudes. A cumulative daily relative
308 risk comparing 11 °C with 25 °C is not a monthly count ratio per five official cold days, and
309 the present event is a first hospitalisation after a first HF diagnosis without recorded admission
310 cause. The CHD hot-night residual should likewise be read against Guo et al. rather than as a
311 replication of it [17]. That study reported no overall association for the official hot-night flag
312 after adjustment for mean temperature, and a positive association for hourly nighttime excess
313 heat [17]. The present analysis uses monthly official counts, a first-event CHD series, and a
314 later decade. A difference between monthly mean temperature and monthly official hot-night
315 counts does not identify an intensity mechanism.

316 Liu et al. (2020, 2026) remain complementary mortality baselines [12,13]. Their
317 attributable fractions and excess-death totals cannot be rescaled into the present count ratios.
318 The failed daily-recovery calibration is the corresponding methods limit: monthly sums do not
319 automatically yield daily trigger estimates [15,16]. That refusal is specific to this series, this
320 implementation, and this calibration standard. It is not evidence that recovery of daily effects
321 from aggregated outcomes fails in general.

322 Pollution and influenza are scientifically motivated in this setting [1,18]. Archive
323 models that entered nitrogen dioxide, fine particulate matter, or ozone used a general-
324 population offset and joint maximum and minimum temperature, so they are not adjusted
325 versions of the separate core panel (Supplementary Table S9). On a monthly grain they cannot
326 separate confounding from mediation for ozone, and no staged estimate was used to revise
327 Table 2. On the 121 months with influenza data, an archive model associated higher influenza
328 activity with higher CHD counts (Supplementary Table S7); that model is not an adjusted
329 version of the core panel and does not adjust Table 2. Confounding by infection, ozone, or
330 nitrogen dioxide is therefore unresolved. Absent cohort person-time, even a stable count ratio
331 remains a count ratio [14].

332 The contribution of this analysis is identification rather than estimation. A monthly aggregate
333 series of first events can be reported in a way that shows where its information comes from.
334 Four displays do that work here: which months carry the exposure contrast, whether the
335 outcome decline is compatible with risk-set depletion, how far each interval moves across
336 standard-error constructions, and which specifications each estimate depends on. A protected
337 primary claim would have required a predeclared confirmatory contrast, multiplicity control
338 that survives the twelve-contrast family, uncertainty constructions that are not chosen for null
339 exclusion, and residual diagnostics that leave no substantial CHD serial correlation
340 unaddressed. This analysis does not meet that bar. We report the complete panel and the
341 refusals.

342

343 **Conclusion**

344 Between 2013 and 2023, hot nights in Hong Kong increased while cold days persisted.
345 In governed monthly aggregates for people with type 2 diabetes and/or hypertension, CHD first
346 hospitalisations were more closely associated with hot nights, and HF first hospitalisations with
347 cold days, than with the other thermal encodings examined. Neither association survived
348 correction across the twelve comparisons. The CHD estimate additionally depended on the
349 treatment of uncertainty and on the inclusion of 2020–2023. This analysis therefore contributes
350 a set of hypotheses, and an explicit account of what monthly aggregate counts cannot settle.
351 Better-denominated and more finely resolved data are required before a thermal effect on
352 cardiac hospitalisation in this cohort can be estimated.

353

354 **Strengths and limitations**

355 **Strengths.** The twelve-contrast panel is reported in full, rather than filtered to its largest
356 estimates. Uncertainty is shown as an explicit four-construction ladder rather than as a single

interval. Extreme-day exposures use published official thresholds. Identification displays show the sources of cold-day variation, the first-event decline, and the dependence of each estimate on the analysis window. The daily-recovery analysis is reported as a refusal rather than as a coefficient.

Limitations. Admission cause was not recorded, so an event is a first hospitalisation after a first diagnosis and not a cardiac-caused admission. Inpatient-only semantics are inferred from the delivered column labels and are not yet confirmed. Monthly counts of cohort members still at risk of a first event were unavailable, so the estimates are count ratios rather than incidence-rate ratios [14]. The design is ecological and monthly, so individual-level and daily-triggering interpretations are not identified [14]. Age, sex, and disease-subtype strata were not delivered. Residual serial correlation remains material for CHD, with lag-1 Pearson autocorrelation of 0.508 in the hot-night model. Official cold days are concentrated in December–February, and only 29 of 132 months carry any official cold day, so the HF cold-day estimate rests on differences between winters. The CHD hot-night estimate is not stable across analysis windows, and is compatible with 1 before 2020. Official hot-night counts are not hourly nighttime excess heat [17]. All exposures were measured at a single Observatory station and applied territory-wide, and within-territory exposure variation was not modelled. Reference rules for monthly hot-tail and cold-tail indicators are not yet fixed, and no such indicator was used as a confirmatory exposure. Confounding by pollution, humidity, and influenza is unresolved in the core panel. A stroke series was named in correspondence but was not delivered, and no stroke result is reported. External dissemination of the governed aggregates requires confirmation from the outcome and supervising investigators.

379

380 Declaration on the use of AI

AI-assisted tools were used for code scaffolding, methodological brainstorming, and review of potential data and modelling issues. Scientific claims, numerical results, and collaborator-owned Methods text were checked against governed outputs and the live weather paragraph.

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385 Declaration of competing interest

The authors declare no competing interests.

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388 Acknowledgements

389 Data and code availability

A disclosure-minimised release of tables, figures, and machine-readable checks accompanies the analysis. Source monthly health counts and merged panels are governed and are not redistributed. External submission requires team confirmation.

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Commented [KW19]: You do not include the co-authors here. The acknowledgement section is for acknowledging people whose contribution helped the study but do not qualify as intellectual contribution. For example if somehow helped you to download the data and just passed it to you for analysis (without any input on direction or commenting), then you would put their name down here. At the moment, I don't think we have anyone to acknowledge.

Commented [KW20]: This needs to be rewritten. All you need to say is that the code used to generate the findings can be found on your github page (provide link). For the data, we need to check with Roro and look at the IRB app to see as to what can be publicly shared.

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Commented [H21]: Changed from Yang C-Y to Yang CY for consistent formatting.

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